Academic Year 2024/2025

Computer Science Department
Sciences Faculty
University of Algiers 1
Information Systems (L3)
Artificial Intelligence

Lab Series #1: Getting Started with Prolog – Sample solution

Objectives
— Learn how to define facts, rules, and run basic queries.
— Use arithmetic operators, conditionals, and list processing in Prolog.

Exercise 1. Robots in Space

In a space base, different robots perform specific functions.

- R1 is an exploration robot.
- R2 is a scientific robot.
- All scientific robots are autonomous.
- All exploration robots are robust.
- An autonomous robot can perform analyses.
- A robust robot can withstand extreme conditions.
- Any robot that performs analyses consumes energy.
- Any robot that withstands extreme conditions consumes fuel.

Question 1. Establish the knowledge base.
Question 2. Perform the following queries:

1. Which robots consume what?
2. List all autonomous robots?
3. What does robot `r1` consume?
4. Find the robots that are both autonomous and robust.
5. Check if a robot consumes both energy and fuel

Exercise 2. Medical Diagnosis Logic

Question 1. Define the knowledge base.

- People who have flu should take paracetamol.
- People who have a fever and are coughing are considered to have flu.
- Anyone with a temperature above 38°C has a fever.
- Ali is coughing and has a temperature of 39°C.

Question 2. Perform the following queries:

1. Should Ali take paracetamol?
2. if the base has multiple individuals, how to check all people who need Paracetamol?
3. Does Ali have a fever?

4. Who has the flu?
5. Does a person with a temperature of 39°C have a fever?

Exercise 3. Animal Kingdom

You are given a classification of animals (see Figure 1).

```
              Animal
                |
         -----------------
         |                |
       Mammal            Bird
         |                |
      --------         --------
      |      |         |      |
     Dog  Elephant   Eagle  Parrot
```

Figure 1. Animal Kingdom.

Question 1. Create rules in such a way that you can answer the queries in Question 2.

Question 2. Answer the following queries using your knowledge base:

- Which animals lay eggs and which give birth?
- Which animals have wings and lay eggs?
- Which animal is a mammal and does not give birth? (*Use the logical negation \+ to test the absence of a fact.*)
- Is there an animal that is a bird but does not lay eggs?

Exercise 4. Student Scores

You are given the following facts:

```
student_score(ali, 15).
student_score(lina, 9).
student_score(samir, 12).
student_score(salma, 18).
student_score(karim, 7).
```

1. Define a rule to determine who passed (score ≥ 10).
2. Write a predicate to list all students who failed.
3. Write a predicate to classify performance (excellent, good, and weak).
4. Define a rule to find the **top scorer (**use the useful predicate: aggregate)**.**
5. write rules to compute the minimum score.
6. Define a rule to calculate the **average score** of all students.
7. write rules to compute the number of students above average.
8. Rewrite the solutions for queries 4 to 7 without using the aggregate predicate.

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